

KAZALY, Kazakhstan

WMO: 358490

Lat: 45.7556N

Lon: 62.1067E

Elev: 68

StdP: 100.51

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.7	-21.9	-28.8	0.3	-24.0	-25.8	0.4	-21.1	12.3	-9.3	10.1	-6.5	1.6	50	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	37.4	20.5	35.8	20.0	34.4	19.4	21.8	34.2	21.0	33.4	20.2	32.1	2.1	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.8	27.0	16.6	11.9	26.3	15.6	11.2	26.1	64.1	34.1	61.0	33.6	58.4	32.0	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	7.3	6.0	-27.4	40.0	3.9	1.8	-30.2	41.3	-32.5	42.3	-34.6	43.3	-37.5	44.6		
(5)			-27.6	23.7	3.8	1.6	-30.3	24.8	-32.6	25.8	-34.7	26.7	-37.5	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-9.3	-7.3	2.9	12.4	19.9	24.8	26.5	24.8	17.9	9.3	0.9	-6.3
	DBStd	13.77	6.68	7.25	6.17	5.51	4.08	3.52	3.15	3.76	4.43	4.80	5.52	6.06
	HDD10.0	2208	599	484	230	38	1	0	0	0	2	73	275	507
	HDD18.3	3907	857	718	479	189	31	2	0	3	59	282	522	765
	CDD10.0	2137	0	0	10	110	308	444	513	460	239	50	3	0
	CDD18.3	794	0	0	0	11	79	197	254	205	47	1	0	0
	CDH23.3	10040	0	0	6	148	973	2547	3315	2466	559	27	1	0
CDH26.7	5042	0	0	1	46	375	1317	1806	1288	206	3	0	0	
Wind	WSAvg	2.1	2.0	2.2	2.5	2.4	2.6	2.1	1.9	1.9	1.8	1.9	1.8	2.0
Precipitation	PrecAvg	129	12	9	15	19	12	7	5	4	9	13	14	13
	PrecMax	200	39	33	63	89	42	26	34	28	59	54	42	52
	PrecMin	36	0	0	1	0	0	0	0	0	0	0	0	0
	PrecStd	38	9	7	11	19	12	8	7	7	13	12	10	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	11.1	22.2	31.1	34.9	38.7	39.9	38.7	34.9	26.6	17.9	6.8
		MCWB	3.0	6.9	12.3	17.3	18.1	20.6	21.6	20.8	18.1	14.2	11.4	3.3
	2%	DB	2.3	6.1	17.7	27.5	32.6	36.6	37.5	36.6	31.3	23.4	13.5	4.5
		MCWB	1.3	3.6	10.4	15.5	17.2	20.0	20.9	20.1	17.4	12.4	8.6	2.8
	5%	DB	1.0	3.5	14.8	24.4	30.7	34.9	35.9	35.0	29.0	20.8	10.9	2.7
		MCWB	0.0	1.5	8.7	13.7	16.6	19.5	20.5	19.6	16.3	11.5	7.0	1.1
	10%	DB	-0.2	1.5	11.7	21.8	28.6	33.3	34.3	33.2	26.6	18.1	8.4	1.1
		MCWB	-1.2	0.1	6.9	12.5	15.9	18.8	19.8	18.9	15.3	10.3	5.2	-0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	8.0	13.1	18.7	19.7	22.8	23.2	22.8	19.6	15.0	11.9	4.5
		MCDB	4.4	10.5	21.2	28.6	30.8	34.9	35.2	35.5	32.2	23.3	16.5	5.6
	2%	WB	1.5	4.1	10.7	16.1	18.4	21.4	22.0	21.2	18.0	13.4	9.3	2.8
		MCDB	2.2	6.3	16.6	24.7	29.4	33.6	34.6	33.8	30.2	21.7	12.8	4.3
	5%	WB	0.1	1.7	9.1	14.5	17.5	20.3	21.2	20.2	16.8	12.1	7.2	1.3
		MCDB	0.9	3.3	14.1	23.1	28.2	32.2	33.6	32.8	27.4	19.2	10.4	2.6
	10%	WB	-1.2	0.2	7.3	13.1	16.6	19.5	20.5	19.5	15.6	10.8	5.6	-0.1
		MCDB	-0.3	1.4	11.5	20.6	26.9	30.9	32.4	31.5	25.6	17.4	8.1	1.0
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.9	10.3	12.0	13.1	13.3	13.0	13.5	13.7	12.8	9.1	7.2
		MCDBR	6.0	9.5	14.5	15.5	15.4	15.1	14.9	15.5	16.2	16.3	12.1	7.9
		MCWBR	5.3	7.0	8.4	6.9	5.6	5.5	5.3	5.8	7.0	8.5	7.8	6.4
	5% WB	MCDBR	5.5	8.8	13.4	14.1	13.9	13.6	13.6	14.2	15.2	14.6	11.2	7.4
		MCWBR	5.0	6.8	8.4	6.8	5.8	5.5	5.3	5.9	7.0	8.4	7.6	6.2
Clear-Sky Solar Irradiance	taub		0.335	0.361	0.430	0.468	0.466	0.446	0.428	0.414	0.380	0.382	0.363	0.342
	taud		2.083	2.049	1.982	1.929	1.998	2.117	2.193	2.208	2.246	2.154	2.145	2.103
	Ebn at Noon		723	785	780	788	807	825	836	833	829	755	686	664
	Edh at Noon		98	124	154	177	171	153	140	133	117	111	93	87
All-Sky Solar Radiation	RadAvg		1.27	2.27	3.60	5.19	6.66	7.25	6.99	6.29	4.87	3.07	1.68	1.03
	RadStd		0.23	0.40	0.32	0.36	0.36	0.39	0.29	0.27	0.18	0.19	0.17	0.17

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	-0.73	N/A	N/A	+146
		N/A	N/A	N/A	N/A	N/A	-0.81	N/A	N/A	+127

Nomenclature: See separate page